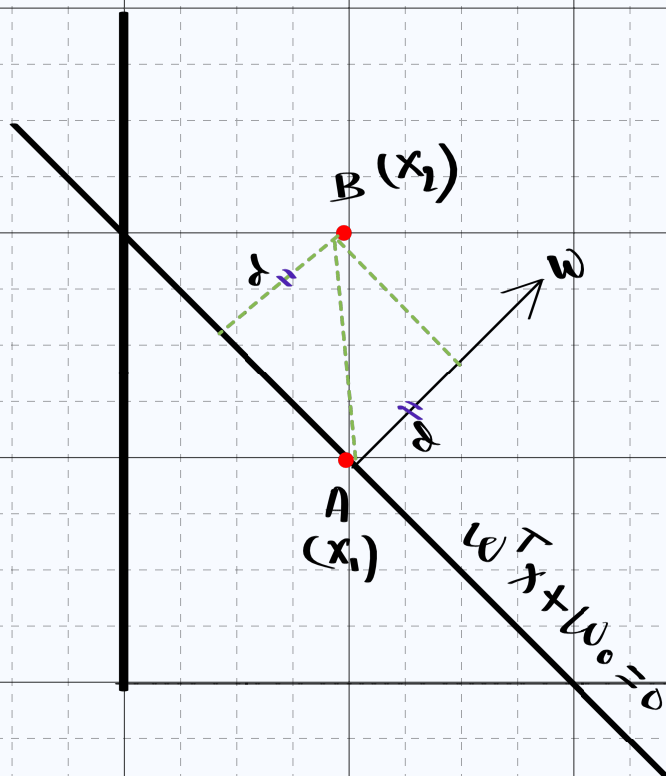


→ Perpendicular distance b/w a point and plane



→ how can I calculate d ?
given - $\vec{AB}$ & $\vec{w}$

$$\begin{aligned} d &= \text{Projection of } \vec{AB} \text{ on } \vec{w} \\ &= \frac{\vec{w}^T \vec{AB}}{\|\vec{w}\|} \end{aligned}$$

$$= \frac{\vec{w}^T (\vec{x}_2 - \vec{x}_1)}{\|\vec{w}\|}$$

$$= \frac{\vec{w}^T \vec{x}_2 - \vec{w}^T \vec{x}_1}{\|\vec{w}\|}$$

$$= \frac{(\vec{w}^T \vec{x}_2 + w_0) - (\vec{w}^T \vec{x}_1 + w_0)}{\|\vec{w}\|}$$

$$= \frac{(\vec{w}^T \vec{x}_2 + w_0) - 0}{\|\vec{w}\|}$$

$\left\{ \begin{array}{l} \because \vec{x}_1 \\ \text{lies on the line} \\ \vec{w}^T \vec{x} + w_0 = 0 \end{array} \right.$

$$\boxed{d = \left| \frac{\vec{w}^T \vec{x}_2 + w_0}{\|\vec{w}\|} \right|}$$